

# Oxidation Numbers and Redox Reactions

Oxidation numbers, agents, half-equations and balancing in acid

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

## RULES AND STEPS

Element: 0 · Monatomic ion: its charge · F: -1 · H: +1 (-1 in metal hydrides) · O: -2 (-1 in peroxides) · Sum = overall charge

Balancing in acid: atoms → O with  $\text{H}_2\text{O}$  → H with  $\text{H}^+$  → charge with  $\text{e}^-$  → equalize electrons → add → check

Check both atoms and charge in every final equation.

## A. Understanding

1. Define oxidation and reduction in terms of electrons and of oxidation numbers.

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## B. Oxidation numbers

2. Give the oxidation number of the named element: (a) N in  $\text{NH}_3$  (b) N in  $\text{HNO}_3$  (c) Cr in  $\text{Cr}_2\text{O}_7^{2-}$  (d) S in  $\text{SO}_4^{2-}$  (e) Fe in  $\text{Fe}_2\text{O}_3$  (f) O in  $\text{H}_2\text{O}_2$  (g) C in  $\text{CO}_2$  (h) Cl in  $\text{ClO}_3^-$

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### C. Agents and half-equations

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3. In  $\text{Cl}_2 + 2\text{Br}^- \longrightarrow 2\text{Cl}^- + \text{Br}_2$ , identify the species oxidized, the species reduced, the oxidizing agent and the reducing agent.

4. Write the two half-equations for  $\text{Mg} + 2\text{H}^+ \longrightarrow \text{Mg}^{2+} + \text{H}_2$  and label each as oxidation or reduction.

5. Which of these are redox reactions? Explain. (a)  $\text{NaOH} + \text{HCl} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$  (b)  $2\text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{O}_2$  (c)  $\text{AgNO}_3 + \text{NaCl} \longrightarrow \text{AgCl} + \text{NaNO}_3$

### D. Balancing

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6. Balance:  $\text{Cu} + \text{Ag}^+ \longrightarrow \text{Cu}^{2+} + \text{Ag}$

7. Balance in acidic solution:  $\text{Cr}_2\text{O}_7^{2-} + \text{Fe}^{2+} \longrightarrow \text{Cr}^{3+} + \text{Fe}^{3+}$

8. Balance in acidic solution:  $\text{Cu} + \text{NO}_3^- \longrightarrow \text{Cu}^{2+} + \text{NO}$

## E. Challenge

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9. (a) Balance in acidic solution:  $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} \longrightarrow \text{Mn}^{2+} + \text{CO}_2$ . (b) In a titration, 25.00 mL of an  $\text{Fe}^{2+}$  solution needs 20.00 mL of 0.0200 M  $\text{KMnO}_4$  to react completely ( $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{3+}$ ). Calculate the concentration of  $\text{Fe}^{2+}$ .

10. Balance in basic solution:  $\text{Cr}(\text{OH})_3 + \text{ClO}^- \longrightarrow \text{CrO}_4^{2-} + \text{Cl}^-$